Tanay Pratap Singh

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Education

Master's of Science in Applied Data Science	**Syracuse University, Syracuse, NY**	Jan. 2025 - Dec. 2026
B.Tech in Computer Science and Engineering	**JSSATE Noida, UP, India**	Dec. 2020 - May 2024

Availability: Graduating Dec. 2026, available Jan. 2027. Eligible for 3 years of US work authorization (OPT and STEM OPT) with no employer sponsorship required.

Research Experience

Graduate Research Assistant — Feb. 2026 - Present

Syracuse University iSchool, Gravity Spy 2.0 (NSF-funded LIGO project) — *Syracuse, NY*

- Reduced Expected Calibration Error from 0.10 to 0.05 on a deep classifier using label smoothing, AdamW, and AUC-based early stopping, ensuring predicted probabilities were reliable for downstream consumers rather than only correctly ranked.
- Trained two-input Convolutional Neural Network with shared MobileNetV2 backbone in TensorFlow/Keras to predict volunteer similarity labels from paired spectrograms; achieved validation AUC of 0.89 on a held-out 2,000-subject set.
- Developed data ingestion pipeline in Python and Pandas against the Zooniverse Panoptes API to parse 276,000+ classifications, reconciling 5 undocumented metadata schemas across 25,104 subjects and 8,293 auxiliary channels.
- Engineered OpenCV preprocessing workflow retrieving 49,000 spectrograms via batch API calls, transforming 1200x1200 mosaics into paired 224x224 heatmaps; built glitch-type by channel similarity matrix to link detector subsystems to failure morphologies.

Selected Projects

ShopFloor AI | *Python, scikit-learn, MLflow, Kafka, Airflow, Docker, FastAPI, LangChain* — Jul. 2026

- Built predictive maintenance system on 10,000 sensor records across 5 failure modes, engineering 16 features including thermal differential and overstrain; best model (Gradient Boosting) reached 0.85 F1 and 0.96 AUC-ROC.
- Deployed 7-service containerized stack with MLflow model registry promotion, an Airflow DAG performing automated daily retraining behind data quality gates, FastAPI serving, and a LangChain plus ChromaDB retrieval assistant over technical documentation.

ZTF Transient Classifier | *Python, Kafka, Avro, dbt, TensorFlow, Airflow, BigQuery, GitHub Actions* — Jul. 2026

- Engineered streaming pipeline ingesting real-time astronomical alerts via Kafka and Avro, processing 27,000 nightly records; served a multimodal CNN combining image cutouts with a tabular feature branch through a FastAPI REST API.
- Monitored production model drift using Population Stability Index; built a tested and documented dbt layer of 8 SQL models and established CI/CD in GitHub Actions with Docker, keeping the warehouse portable across DuckDB, BigQuery, and Snowflake.

Determinants of Child Mortality: A Cross-National Study | *R, Bayesian Inference, ANOVA* — Jan. - May 2025

- Modeled child mortality across 180+ countries using multiple regression, ANOVA, and Bayesian hypothesis testing, establishing that educational attainment explains as much variance as national income (Bayes Factor 1.65e288 against the null).

Technical Skills

Languages: Python, SQL, R, JavaScript, HTML/CSS

Machine Learning: Scikit-learn, TensorFlow, Keras, XGBoost, CatBoost, LightGBM, SHAP, OpenCV, LangChain; Gradient Boosting, Random Forest, SVM, CNN, LSTM, NLP, Deep Learning, Ensemble Methods, Feature Engineering, Cross-Validation, Model Calibration, Model Evaluation, Model Monitoring, Drift Detection, RAG, LLM

Statistics and Experimentation: Probability, Hypothesis Testing, Experimental Design, A/B Testing, Bayesian Inference, ANOVA, Linear and Logistic Regression, PCA, Time Series Forecasting, Anomaly Detection, Validation Analysis

Data and MLOps: Pandas, NumPy, SciPy, Spark, Docker, Apache Kafka, Apache Airflow, MLflow, Model Registry, dbt, Snowflake, BigQuery, DuckDB, MySQL, ETL, Data Pipelines, Streaming Data, CI/CD, GitHub Actions, Git, GCP, Azure, FastAPI, Tableau, Power BI